

Project: Nyc Urban Mobility
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Urban Mobility Data Summary Report

Urban Mobility Data Explorer is an end-to-end analytics system built to analyze 7.6 million New York City taxi trip records from January 2019. The project processes large-scale mobility data to identify temporal, spatial, and revenue patterns. After cleaning, 7,527,445 records were retained, achieving a 98.2 percent retention rate. The system includes a structured data pipeline, a RESTful API with 12 endpoints averaging under 100 milliseconds response time, and an interactive dashboard with five analytical views.

System Architecture:

- Backend: Flask (Python) with SQLite database.
- Frontend: HTML5, CSS3, Vanilla JavaScript.
- Visualization: Chart.js and Leaflet.js.
- Data processing: Pandas, GeoPandas, PyProj.
- Normalized schema with indexed columns reduced query time from ~2 seconds to under 100 ms.
- Eight derived features (duration, speed, fare metrics, time indicators) computed during ETL for performance and consistency.

Data Pipeline:

- Shapefile to GeoJSON conversion with coordinate reprojection.
- Data integration using chunk processing (100K rows per batch).
- Cleaning removed 140,347 outliers (fares, distances, durations, invalid values).
- Final structured SQLite database (1.2 GB) with optimized indexing.

Algorithms Implemented:

- Custom hash table for $O(1)$ average lookups.
- Merge sort for $O(n \log n)$ stable sorting.
- Binary search for $O(\log n)$ retrieval.
- Manual group-by aggregation in $O(n)$ time.

Key Insights:

- Peak demand: 6–7 PM (45K+ trips/hour); lowest: 4–5 AM (<5K/hour).
- Weekday trips are 23% higher than weekends.
- Manhattan accounts for 68% of trips.
- Average fare: \$13.26; average tip: 16.8%.
- Weekend fares 12% higher; peak revenue 7–9 PM (\$180K+/hour).
- Average trip: 2.8 miles, 15.3 minutes, 11.2 mph (urban congestion indicator).

Project Impact:

- Demonstrates full-stack development, large-scale data processing, database optimization, and algorithm implementation.